Lesson notes · for you

Adversaries and noise

University, vision 4

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Rebuild from a code
- Fool a discriminator
- Remove noise step by step

Activity: In the lab

1. Set the noise near the middle. Compare the two pictures and the difference.
2. Raise the noise to 1, at the far right, and read the difference again.
3. Write what one step did not restore.

Discussion

- Which failure is whose?

Check yourself

- Name the method that undoes noise.

Try it this week
Write the three methods and one failure each, from memory.

Worksheet

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Name _____ Class _____ Date _____

1 **In a GAN, what does the discriminator do?**

- ◯ Invents new pictures
- ◯ Removes a little noise at a time
- ◯ Judges whether a picture is real or made up

2 **A picture from a generator is evidence of what it shows.**

- ◯ True
- ◯ False

3 **Match the generator.**

Autoencoder	• • • •	Undo noise one step at a time
GAN	• • • •	Rebuild from a short code
Diffusion	• • • •	A generator tries to fool a discriminator

4 **Order a diffusion story.**

- ☐ Repeat the step
- ☐ Add noise until the picture is gone
- ☐ Learn a step that removes a little

✎ **Why is one denoise step not a picture coming back?**

Check your answers

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1 **Judges whether a picture is real or made up**

The generator invents and the discriminator judges.

2 **False**

It is a sample from a trained model, not a record of something that happened.

3 **Autoencoder — Rebuild from a short code**
GAN — A generator tries to fool a discriminator
Diffusion — Undo noise one step at a time

Three different losses. The lab only shows one denoise step.

4 **Top to bottom: 3, 1, 2**

The lab stops after one step, on purpose.